Covering note: your 2026-10-15 UID 392 remaining 3 MUSTs are landed

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Date: 2026-10-16
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CC: Robin Langer (`langer.robin@gmail.com`)

Repo: `github.com/scotmacbeth/work-in-progress` **content commit** `16b051d` **stamp commit** `05f773d`
(two-commit pattern; each PDF page 1 now carries content commit `16b051d`, supersedes `3954636`).

To: Rick (`grandparick20@gmail.com`) **CC:** Robin Langer (always)
Date: 2026-10-15 · WRITE session · re-delivery of three corrected PDFs
Attach on send: the three PDFs under `papers/` listed below.
Review artifact: `proofs/reviews/2026-10-15-uid392-corrections-rick.md` (your UID 228 + UID 229 v2 erratum).

Thank you for the check of the five `3954636` PDFs (UID 228/229). All three remaining MUSTs are applied; no theorem statements changed. Where a *printed* statement was false (Cor. 5), I took your route (b). Attribution SHOULDs for Thm 7 and the two-ends §8 witness are done. Registry node grades are unchanged (the gaps were in printed artifacts; the underlying nodes stay `proved`), and I did **not** self-promote — promotions wait for your re-check.

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1. δ-quadratic — `papers/delta-additive-waist-quadratic.tex` (MUST A, partial → done)

- Added the scope **on** Out^0 to the two unscoped headline places you flagged: §1 (the "χ vanishes exactly when δ does" phrasing is now $\chi|_{\bar{A}} \equiv 0$ **iff** $\delta|_{\bar{A}} \equiv 0$ **for every abelian** $\bar{A} \subseteq \mathrm{Out}^0$) and §7 "Theory" (the "coupling = nonabelian lift = non-additivity" slogan now reads **"on** Out^0**, . . . "**). Thm 6 was already scoped and is untouched.
- **Attribution:** Theorem 7 (common receptacle off Out^0) and the counts 0/10304, 0/36420, 14 of . . . , 50 of . . . are credited to you (computed + by hand, 2026-10-14), at the theorem and in a new footnote-style line. The page-1 provenance sentence no longer says "introduces no new claim beyond the cited proof file" — it now names Thm 7 and the counts as your import.

2. Capstone — `papers/inventory-gerbe-couples-on-torus.tex` (MUST B, route (b))

- **Re-adopted** $k_{g,f} \equiv e$, matching the gerbe and floor notes, and **deleted** the §2 "We do not assume $k_{g,f} \equiv e$" sentence. The dependent §2 "strictly inner-valued α is already a functor . . . k are not what make α functorial" sentence is removed.
- **Def. 4 / Cor. 5 / Rem. (D1-vs-D2) reframed** so the honest floor is separated from the naive one by whether the **stratum-three central lifting datum** is retained — *not* by k. Def. 4 now reads "T strict, under standing $k \equiv e$ (so $\alpha_{gf} = \alpha_g \alpha_f$)"; the Cor. 5 proof sketch

and the D1/D2 remark use the central lifting datum. This removes the T-strict-forces-k-central contradiction, so Cor. 5 is true as printed (your $B(\mathbb{Z}/3)$, D_4 0-vs-32 witness no longer bites). A nontrivial central $k \in Z(K)$ is noted as a separate 2-cocycle shifting the sharp line, excluded here (as in the gerbe note). Underlying lemma with $k \equiv e$ is unchanged; node stays `proved`.

3. Two gerbe ends — `papers/2026-10-13-two-gerbe-ends-forced-correlatio` (MUST C, new)

- **Rewrote the false §6 "Scope honesty" nilpotency sentence.** Now: SD_{16}, D_{16} (both order 16) are nilpotent (D_{2^k} has class $k-1$); the wider family D_n, $4 \mid n$, $n > 4$, is **not** uniformly nilpotent — D_{12}, D_{20} (orders 12, 20) are listed as **non-nilpotent rows** that fall outside Rem. (nilpotent extension) yet still compute (δ no, $[\mathfrak{g}]$ yes) and so *happen* to satisfy the implication. Dropped the "class 3 / nilpotent" claim for the whole family.
- **§8 centre-size fix:** the forbidden corner is now keyed off **"no nontrivial commutator is central"**, not "$Z(K)$ trivial" — with your $K = \mathrm{Dih}(C_3^2) \times C_2$ ($Z = C_2 \neq 1$, still in the corner) named as the reason centre size is not the parameter.
- **Attribution:** the $\mathrm{Dih}(C_3^2)$ witness, the "12 unordered Out-pairs" count, and the S_3, A_4 controls are credited to you (2026-10-14).
- **NIT:** the §-Verification "Remark … below" cross-ref corrected to "above".

Per-paper status after this round (your §5 verdict, as I read it)

Paper	Status
δ-quadratic	M3 landed → should move to *checked-sober* on your re-check
Capstone (couples on torus)	Cor. 5 now true as printed (route (b)) → one-round-closed
Two gerbe ends	N1 landed → ready
Supply-chain inventory gerbe	already ready (unchanged this round)
Composition is a nerve	already ready (unchanged this round)

Open **SHOULD/tone** items I did *not* touch this round (not MUSTs): the two-ends title/abstract still say "strict one-way forced correlation" without flagging that the central end only detects nonabelianness; and your 2026-10-14 gerbe/nerve SHOULD/NIT list (SHOULD-1/-2/-3, the D-clash, W1/W2/W5) was out of scope here. Say the word and I'll fold the title-tone note in next round.

— MacBeth